

Adaptive Noise-Resistant Threshold Estimation for Quantum Zero-Knowledge Authentication

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Abstract As quantum communications advance, reliable user authentication becomes essential. This work studies a quantum identity authentication protocol under realistic noise, deriving a suitable threshold for Quantum Bit Error Rate verification and proposing an adaptive strategy that improves robustness and correctly addresses information leakage across repeated authentication attempts.

Introduction

Since the BB84 protocol [1] was introduced, Quantum Key Distribution (QKD) has progressed from theoretical proposals to practical implementations. These schemes have encryption as their goal, while less attention has been devoted to authentication. Quantum Identity Authentication (QIA) fulfills this need through information-theoretic security; this extra layer is needed since quantum channels ensure confidentiality and integrity, but don't natively verify user identities, which is needed in scenarios such as shared quantum terminals and multi-factor authentication systems. Authentication is closely related to Zero-Knowledge Proofs (ZKPs), where a prover demonstrates knowledge of a secret without revealing it [2]. Inspired by this, a quantum ZKP-based authentication protocol has been proposed [3]: Alice and Bob derive bitstrings from a shared secret to define state preparation, measurement bases, and encryption keys for the outcomes. Alice then sends Bob a photon stream, receives back an encrypted subset of measured states δ_b , and computes the Quantum Bit Error Rate (QBER): if it is below a threshold θ_v , Bob is verified.

While the protocol attempts to account for noise through a QBER-based threshold, its choice is not rigorously justified: $\theta_v = 11\%$ is inherited from BB84 as the point at which the Shannon rate vanishes [4]. However, the Quantum ZKP (QZKP) protocol requires a dedicated analysis. An inadequate threshold may lead to rejecting honest users or accepting malicious ones, and the protocol is also vulnerable to key-space reduction attacks.

In this work, we investigate an appropriate QBER threshold for the QZKP protocol introduced above. We analyze security in the multi-session setting, and propose information leakage mitigation strategies, which we validate through simulated protocol runs. The results show high accuracy even in noisier environments, allowing for higher thresholds than the QKD inspired 11% without losing security. The adaptive nature of the noise estimation allows for a threshold tailored to

the physical deployment.

Noise-resistant threshold choices

Differently from the standard application of the BB84 protocol, the QBER calculated in the QZKP protocol is used to determine whether the prover is aware of the shared secret or not. Since a deterministic distinction is not achievable, this is a statistical distinguishability problem. We model the quantum channel as a Binary Symmetric Channel (BSC), where each transmitted bit is flipped independently with probability p , considered as the crossover probability. The observed QBER is $\hat{q} = \frac{k}{N}$, where k is the number of observed errors and N is the length of the communicated subset δ_b .

Authentication can be formulated as a binary hypothesis testing problem, where $\mathcal{H}_0$ indicates that the prover does not know the secret (dishonest case), and $\mathcal{H}_1$ means the prover knows the shared secret (honest case). A dishonest prover attempting to guess the correct measurement result (and its encryption) behaves effectively as a random source. Knowledge of the encryption algorithm alone does not improve the adversary's success probability, since authentication relies on quantum correlations determined by the shared secret, nor does the opposite, since even correct measurements need to be encrypted correctly. Considering all possible replies a dishonest prover could send back to Alice, the probability of correctly reproducing each expected bit is 0.5. Therefore, the dishonest prover induces an effective crossover probability $p_i = 0.5$, which corresponds to a completely random channel BSC(0.5).

For an honest prover, errors arise only from physical imperfections, including channel noise. We denote the resulting error rate as q_h , which depends on system parameters and operating conditions. To distinguish between the two hypotheses, we use the Kullback–Leibler (KL) divergence between Bernoulli distributions:

$$D(a \parallel b) = a \log_2 \frac{a}{b} + (1-a) \log_2 \frac{1-a}{1-b}. \quad (1)$$

In particular, the divergence from the fully random

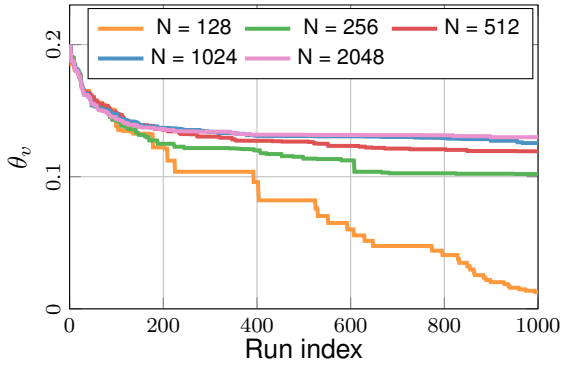
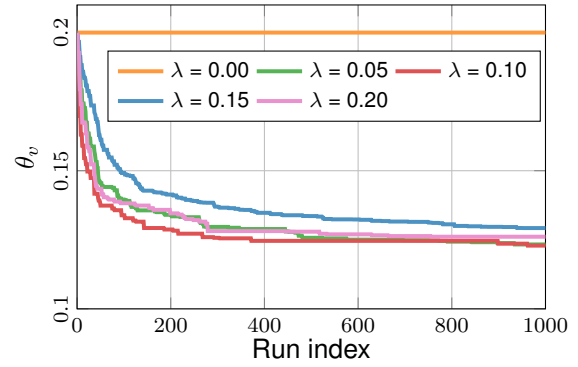
(a) θ_v with varying N (b) θ_v with varying λ

Fig. 1: Adaptive threshold behavior

case satisfies $D(q \| 0.5) = 1 - H_2(q)$, where $H_2(\cdot)$ is the binary entropy function. This highlights that distinguishability from a dishonest prover corresponds to deviation from randomness.

Using large deviation theory, the probabilities of authentication errors decay exponentially with N :

$$P_{FA} \approx 2^{-ND(\theta_v \| 0.5)}, \quad P_{FR} \approx 2^{-ND(\theta_v \| q_h)} \quad (2)$$

where P_{FA} is the false accept probability (accepting a dishonest prover), and P_{FR} is the false reject probability (rejecting an honest prover). Our strategy considers a trade-off between false accept and false reject probabilities, by making their error rates equal and adjusting the false acceptance rate with a tuning parameter $\alpha \in (0, 1)$:

$$D(\theta_v \| q_h) = \alpha D(\theta_v \| 0.5). \quad (3)$$

$$\theta_v = \frac{\alpha \ln((1 - q_h)/(0.5))}{\ln((1 - q_h)/(0.5)) + \ln((0.5)/(q_h))}. \quad (4)$$

This choice allows us to minimize the overall error probability in the asymptotic regime, allowing to give more weight to the false acceptance. In practical implementations, the honest error rate q_h is time-varying due to device imperfections and environmental fluctuations. To account for this, we estimate q_h using a grouped QBER approach [5]. The states in δ_b are partitioned into groups of size b . For each group G_i and its observed errors k_i , the local QBER is computed as $q_i = k_i/b$. We estimate the honest error rate using an average over all groups that satisfy $q_i \leq \theta_v$, to reduce the sensitivity to transient disturbances. To improve stability, we apply an exponential update, introducing a parameter $\lambda \in (0, 1)$ to regulate how fast the threshold adapts:

$$\hat{q}_h^{(i)} \leftarrow (1 - \lambda)\hat{q}_h^{(i-1)} + \lambda \bar{q}_i, \quad 0 < \lambda \leq 1. \quad (5)$$

The estimator is initialized with a conservative default $\theta_v^{(0)} \in [0.01, 0.03]$. At every run, the grouped QBER q_h is calculated and the resulting $\theta_v^{(i)}$ is obtained from 4. To account for Man In The Middle (MITM) attacks, the threshold is only considered if lower than 25% and is never increased. This value is then used in all threshold selection rules, enabling adaptive authentication under slowly varying channel conditions, until the threshold is representative of the system.

Beyond eavesdropping and MITM attacks, the protocol isn't fully ZK compliant and is vulnerable

to information leakage under repeated authentication attempts [6]. Each execution returns a binary outcome (accept/reject) based on a statistical test, which effectively provides the attacker with a noisy oracle on the validity of candidate keys. Over multiple sessions, this can be exploited to progressively eliminate inconsistent candidates, leading to a reduction of the effective key space:

$$|\mathcal{K}_{i+1}| \approx |\mathcal{K}_i| P_{\text{accept}}.$$

With $P_{\text{accept}} \approx 2^{-ND(\theta_v \| 0.5)}$, each round leaks approximately $ND(\theta_v \| 0.5)$ bits of information. Although the decision is probabilistic and affected by noise, the leakage accumulates across sessions when a similar decision rule is reused excessively. Grouped QBER estimation and adaptive thresholds can already mitigate this by varying the decision boundary, reducing key elimination efficiency since the attacker doesn't know how permissive the system is. In practice, limiting the number of protocol runs per secret is more effective.

Simulation results

To find ideal parameters, we adapted a QKD system to gather data from its quantum channel. The data showed a best-case-scenario QBER of around 2.2%. The samples (as shown in Fig. 3) were taken by isolating single-photon detections, and then observing cases where the prover was honest (case 0) and where the prover was dishonest (case 1). To this we added a number of mixed cases, such as different noise levels during the protocol (2-3) and different amounts of MITM (sampled without sifting) interference (4-6). These cases contribute to estimating a much higher noise than the best-case-scenario, making the threshold more noise-resistant. Out of all the cases, only 0 and 2-4 should be accepted, since they either contained regular noise or a very small amount of MITM interference. These cases were randomly sampled over 1000 runs, evaluating the adaptive QBER and updating it only in the case of acceptance. Different values of parameters were explored to find the best adaptive behaviour and tradeoff between False Positive Rate (FPR), False Negative Rate (FNR) and overall accuracy. When selecting N , we can observe in Fig. 1a that a short length of 128 bits led to overfitting even below the

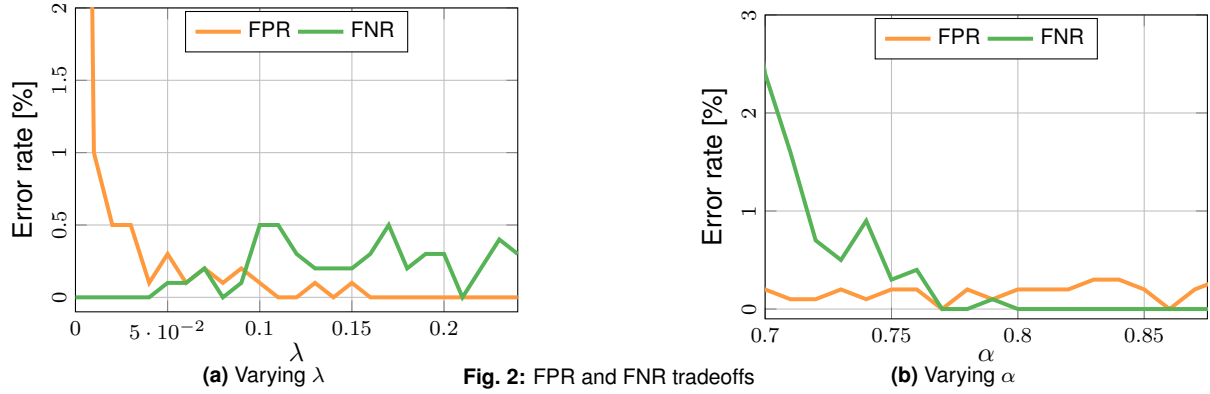


Fig. 2: FPR and FNR tradeoffs

real noise rate. This setting also yielded low accuracy, whereas accuracy was high for $N=1024$. With similar analysis, this length also resulted in the best tradeoff between FNR and FPR, and the ideal group size b was found to be around 50. $\theta_v^{(0)}$ was set to 0.2, which showed good results both in adaptivity and accuracy: the latter could have been even higher starting from a lower $\theta_v^{(0)}$, but it was chosen to trade a few initial false positives for generality. To avoid this, a short calibration phase could be run where we reject anything by design. The value would still safely stay below MITM bounds and allows for unfavorable beginnings regarding noise in less performing setups, nevertheless adapting well to better performances. The adaptive behavior is mostly controlled by λ , whose effect can be seen in Fig. 1b. Varying λ also affected FNR and FPR, whose tradeoff is shown in Fig. 1b. The best accuracy was seen with a value of 0.075, which didn't overfit but achieved a low value in both rates, therefore it is set as the ideal λ . Finally, the balance parameter α had the most effect on FNR/FPR tradeoff, as shown in Fig. 2b, and the best value for high accuracy and low FPR (most important for security) was 0.77. Overall performance was calculated taking all these values into account and sampling on different scenarios, as shown in Fig. 3. For our system and the samples used to run the analysis, after 1000 runs the result was $\theta_v = 0.127$. The protocol achieved an overall 99.6% accuracy, with both FPR and FNR around 0.2%. The relatively low error threshold implies a strong distinguishing capability between correct and incorrect keys, but still remains higher than its counterpart in

BB84, validating our theoretical assumptions. For $\theta_v = 0.127$, the Kullback–Leibler divergence from the uniform error rate 0.5 is $D(0.127 \parallel 0.5) \approx 0.451$ bits. Each authentication round therefore yields approximately $1024 \times 0.451 \approx 462$ bits of statistical information on the validity of possible keys (quantifying their distinguishability rather than directly revealing key bits). Since δ_b is a random encrypted subset of the shared secret, exploiting this information requires aggregation over multiple sessions. The impact depends on the relative size of δ_b and the number of allowed authentication attempts. Reducing N lowers information gain per round but increases false rejection due to reduced statistical reliability, highlighting the trade-off between leakage and accuracy in parameter selection.

Conclusions

This work analyzed a QIA protocol under realistic noise, showing that threshold selection must be tailored to authentication rather than inherited from QKD. An adaptive threshold based on statistical distinguishability achieves high accuracy while remaining robust to fluctuations, but multi-session analysis still highlights a vulnerability to key-space reduction attacks, showing that the protocol isn't fully Zero Knowledge. While the adaptive strategy reduces information leakage, limiting the number of protocol runs per shared secret remains the most effective mitigation. The proposed approach provides a practical balance between security and reliability in noisy quantum environments, leaving for further research a deeper exploration of the theoretical aspects of ZK theory, for example making the protocol mutual rather than one-way.

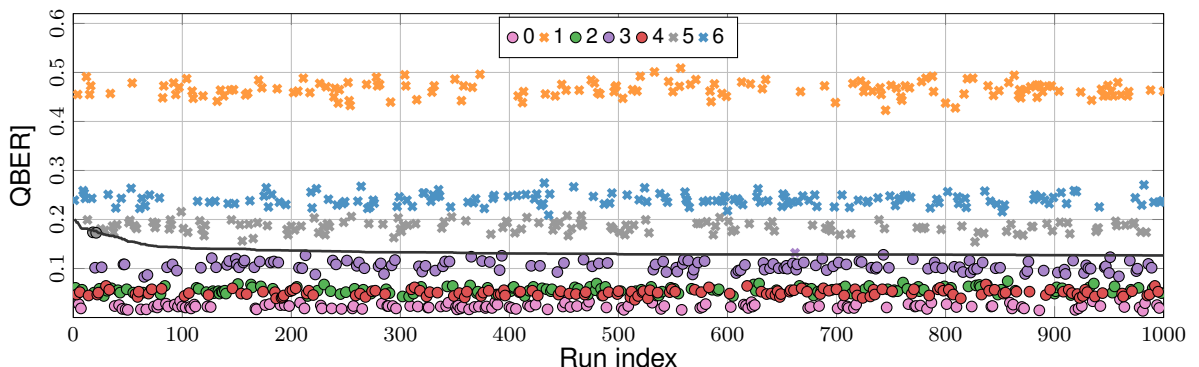


Fig. 3: Overall verification of different sample runs, along with θ_v calculation over runs, and circles showing acceptance versus crosses showing rejections.

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